

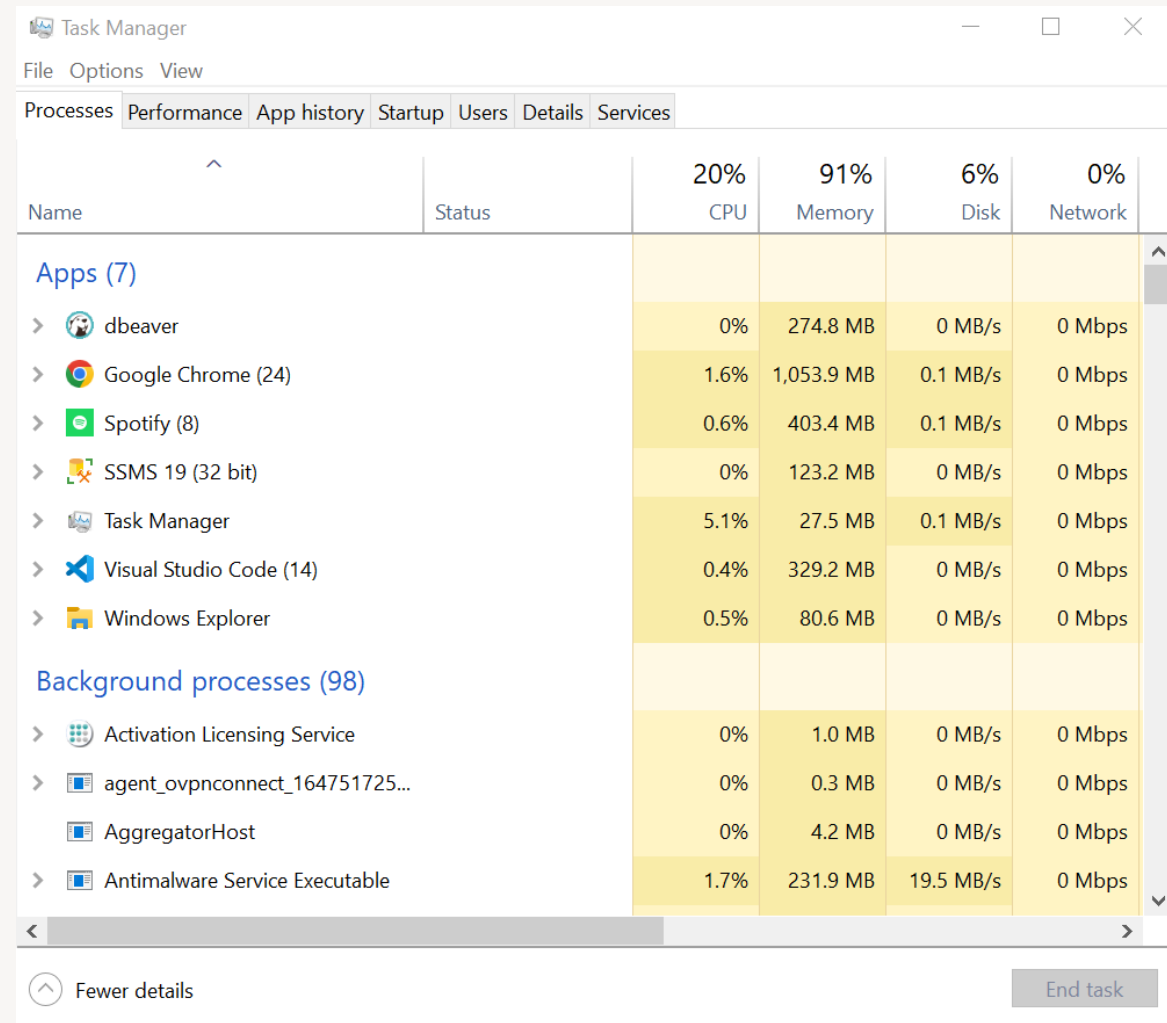
Data Integration using SQL Based MPP Engine

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Limitations



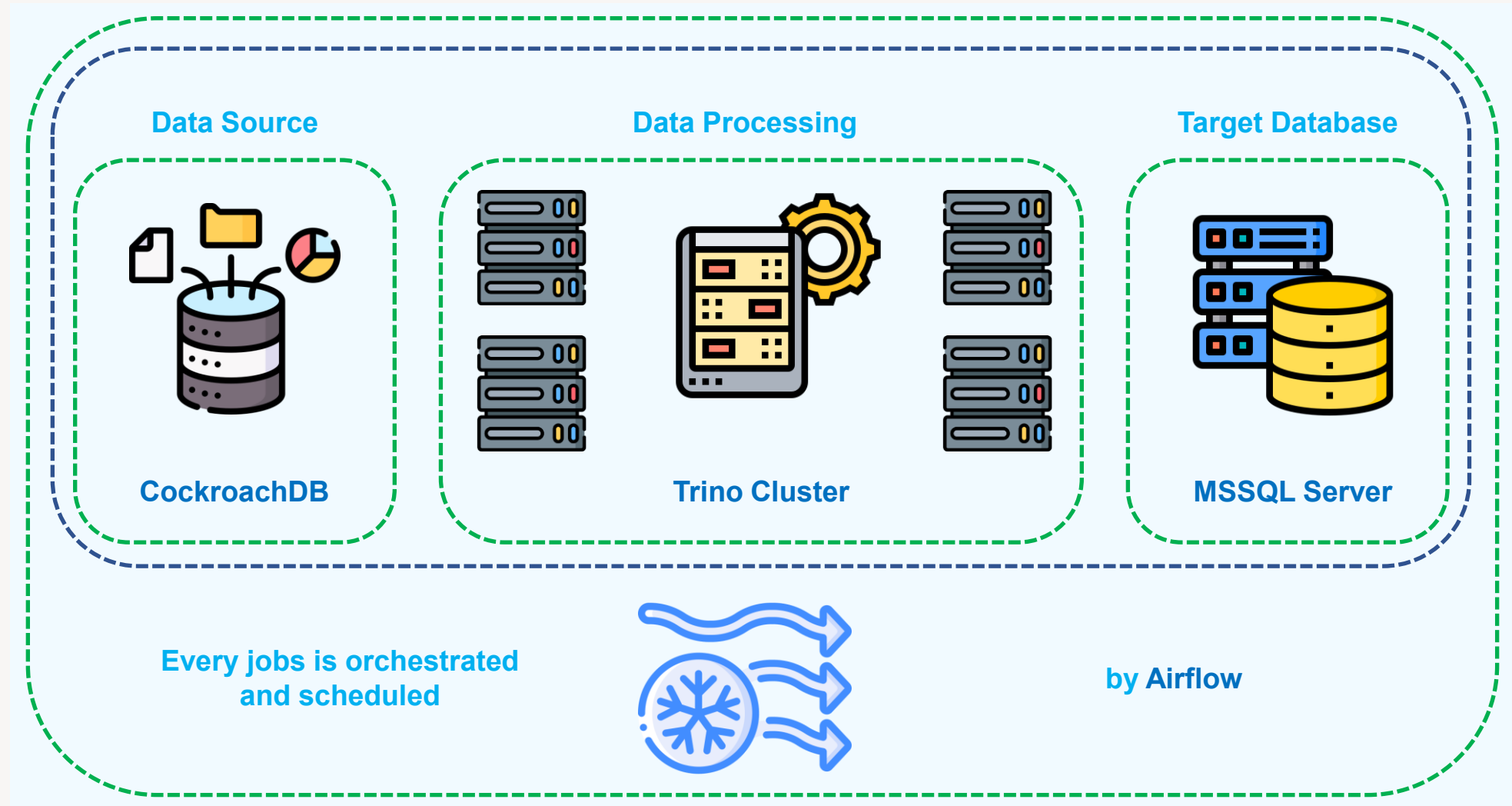
The image shows the Windows Task Manager Performance tab. It displays a table of system resource usage. The 'Apps (7)' section lists several applications with their respective CPU, Memory, Disk, and Network usage. The 'Background processes (98)' section lists system services. The 'End task' button is visible at the bottom right.

Name	Status	20% CPU	91% Memory	6% Disk	0% Network
Apps (7)					
> dbeaver		0%	274.8 MB	0 MB/s	0 Mbps
> Google Chrome (24)		1.6%	1,053.9 MB	0.1 MB/s	0 Mbps
> Spotify (8)		0.6%	403.4 MB	0.1 MB/s	0 Mbps
> SSMS 19 (32 bit)		0%	123.2 MB	0 MB/s	0 Mbps
> Task Manager		5.1%	27.5 MB	0.1 MB/s	0 Mbps
> Visual Studio Code (14)		0.4%	329.2 MB	0 MB/s	0 Mbps
> Windows Explorer		0.5%	80.6 MB	0 MB/s	0 Mbps
Background processes (98)					
> Activation Licensing Service		0%	1.0 MB	0 MB/s	0 Mbps
> agent_ovpnconnect_164751725...		0%	0.3 MB	0 MB/s	0 Mbps
> AggregatorHost		0%	4.2 MB	0 MB/s	0 Mbps
> Antimalware Service Executable		1.7%	231.9 MB	19.5 MB/s	0 Mbps

Resources

- 61 % of my resources is consumed by **WSL**, it is a bottleneck
- In this PoC, I only deploy the environment on single-node cluster

Overall Architecture

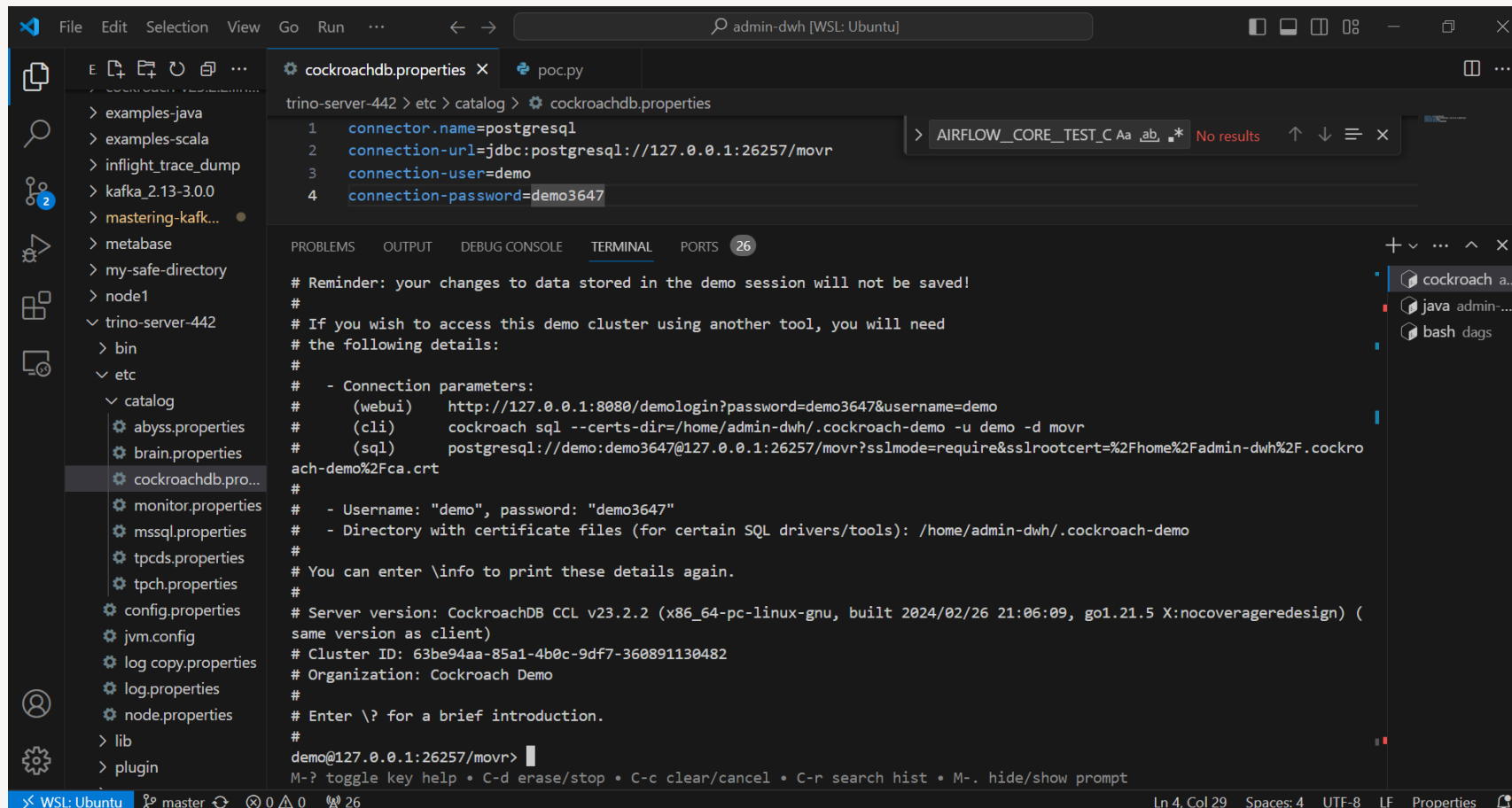


Commands

```
cockroach init
cockroach start-single-node
clear;
cockroach demo
trino-server-442/bin/launcher run
trino-server-442/bin/launcher run
show catalogs;
trino-server-442/bin/launcher run
./trino-server-442/trino.jar
./trino-server-442/trino.jar http://localhost:8081
pip3 install apache-airflow
airflow db init
export AIRFLOW_HOME=~/.airflow
airflow db init
sudo apt install software-properties-common
sudo apt-add-repository universe
sudo apt install python-setuptools
sudo install python3-pip
sudo apt install libmysqlclient-dev
sudo apt install libssl-dev
sudo apt install libkrb5-dev
```

```
mkdir airflow
export AIRFLOW_HOME=~/.airflow
pip3 install apache-airflow
sudo reboot
airflow db init
./trino-server-442/trino.jar http://localhost:8081
restartd airflow webserver -p 8082 --daemon && restartd
airflow scheduler --daemon
sudo apt install x11-session-utils
sudo apt install restartd
restartd airflow webserver -p 8082 --daemon && restartd
airflow scheduler --daemon
stop airflow webserver
ps aux | grep airflow
airflow scheduler --daemon
```

Add CockroachDB Connection to Trino



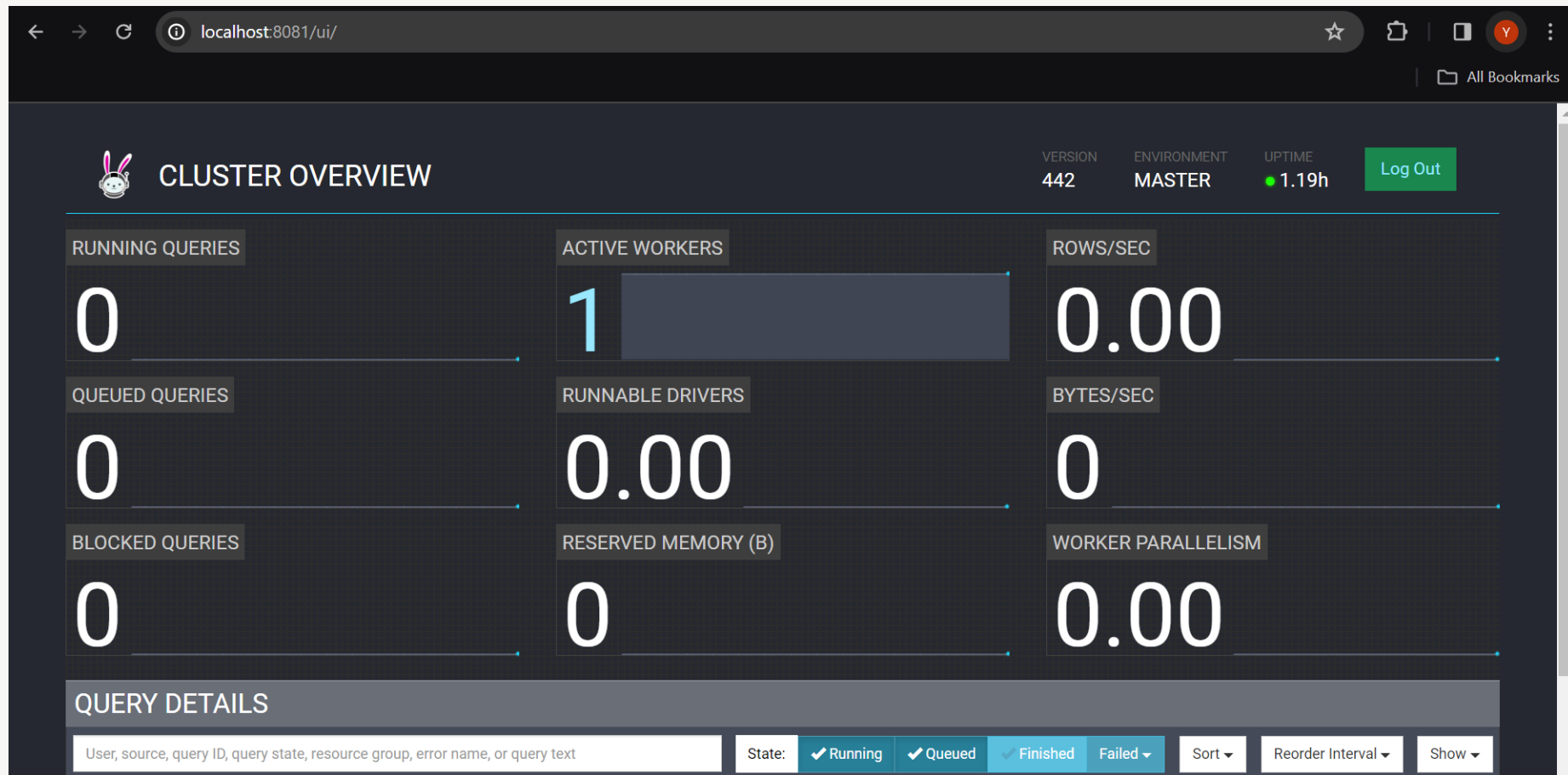
```
File Edit Selection View Go Run ... admin-dwh [WSL: Ubuntu]

cockroachdb.properties X poc.py
trino-server-442 > etc > catalog > cockroachdb.properties
1 connector.name=postgresql
2 connection.url=jdbc:postgresql://127.0.0.1:26257/movr
3 connection.user=demo
4 connection.password=demo3647

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 26

# Reminder: your changes to data stored in the demo session will not be saved!
#
# If you wish to access this demo cluster using another tool, you will need
# the following details:
#
# - Connection parameters:
#   (webui) http://127.0.0.1:8080/demologin?password=demo3647&username=demo
#   (cli)   cockroach sql --certs-dir=/home/admin-dwh/.cockroach-demo -u demo -d movr
#   (sql)   postgresql://demo:demo3647@127.0.0.1:26257/movr?sslmode=require&sslrootcert=%2Fhome%2Fadmin-dwh%2F.cockro
# ach-demo%2Fca.crt
#
# - Username: "demo", password: "demo3647"
# - Directory with certificate files (for certain SQL drivers/tools): /home/admin-dwh/.cockroach-demo
#
# You can enter \info to print these details again.
#
# Server version: CockroachDB CCL v23.2.2 (x86_64-pc-linux-gnu, built 2024/02/26 21:06:09, go1.21.5 X:nocoverageredesign) (
# same version as client)
# Cluster ID: 63be94aa-85a1-4b0c-9df7-360891130482
# Organization: Cockroach Demo
#
# Enter \? for a brief introduction.
#
demo@127.0.0.1:26257/movr>
M-? toggle key help • C-d erase/stop • C-c clear/cancel • C-r search hist • M-. hide/show prompt
```

Implementation



Create Airflow DAG

The image shows a VS Code editor window with a dark theme. The main editor area displays a Python file named `poc.py`, which is part of an Airflow DAG. The code defines a function `_stg_users()` that creates a database engine, connects to a CockroachDB instance, and executes a query to insert data into a staging table. The DAG is configured with a daily schedule and a catchup of False. The left sidebar shows the file explorer with the project structure, including files like `airflow.cfg`, `airflow.db`, and `webserver_config.py`. The bottom status bar indicates the current file is `poc.py` and the editor is in WSL: Ubuntu. The terminal at the bottom shows the command `demo@127.0.0.1:26257/movr>`.

Run DAG in Airflow Web UI

The screenshot displays the Airflow Web UI for the DAG 'daily_trino'. The top navigation bar includes links for DAGs, Cluster Activity, Datasets, Security, Browse, Admin, and Docs. The current time is 05:51 UTC. The DAG is scheduled at 00 12 *** and the next run is on 2024-03-23 at 12:00:00. The 'Grid' view is selected, showing a list of DAG runs. The filter bar shows the date 03/23/2024 05:36:22 AM, 25 runs, and filters for All Run Types and All Run States. The task duration chart on the left shows a single run with a duration of 00:00:04. The DAG runs summary table shows 1 total run displayed, all successful.

DAG Runs Summary	
Total Runs Displayed	1
Total success	1

Store Processing Result In SQL Server

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'TrinoMSSQL' database selected. The right pane shows a query window with the following SQL query:

```
1 SELECT TOP (1000) [city]
2     ,[name]
3     ,[address]
4     ,[credit_card]
5 FROM [TrinoMSSQL].[staging].[stg_users]
6
```

Below the query window, the 'Results' pane shows the execution results of the query. The status bar indicates 'Query executed successfully.' and '50 rows'.

	city	name	address	credit_card
1	amsterdam	Alexander Graham	77701 Hudson Villages	6736266953
2	amsterdam	Alexander Combs	75084 Gabriel Terrace Suite 72	7413163322
3	amsterdam	Philip Grant	7371 Michael Forks Suite 9	9784578052
4	amsterdam	Aaron Bailey	41016 Santiago Brooks	6198911787
5	amsterdam	Damon Dorsey	29007 Best Cliff	5286351400
6	boston	Kimberly Evans	94263 Kerri Parkway	5029681004
7	boston	Elizabeth Cummings DDS	67907 Hayes Plains Suite 89	7389933244
8	boston	James Gregory	24371 Carla Mount Suite 82	0102791087
9	boston	Deborah Sims	18692 Timothy Avenue	5903683621
10	boston	Kristen Clark	13926 Brendan Mountains	1377307817
11	boston	Betty Campbell	32873 Ian Hills	4019906180
12	los angeles	John Taylor	72577 James Ways Apt. 30	8061473501

Thank you

